T06 THERMAL COMFORT MONITORING | O (MAX : 1 PT)

Intent : Monitor and effectively address unacceptable thermal comfort conditions and inform building managers and users of the thermal comfort parameters of their indoor environment.

Summary : This WELL feature requires projects to monitor thermal comfort parameters using sensors in their buildings that can be used as feedback for building managers and users to take appropriate actions.

Issue : Unfavorable levels of heat, humidity and ventilation are associated with people's experience of itchy eyes, headache and throat irritation.[1] Outdoor weather, indoor occupancy and building physics and performance, including ventilation rates, are highly variable and have a direct impact on human perceptions of thermal comfort. To maintain ideal performance metrics, projects should continuously gather data on thermal comfort parameters in order to inform remediation actions.

Solutions : Building HVAC systems should be designed to monitor and control for variations in indoor air temperature, mean radiant temperature, relative humidity and air movement. Thermal comfort monitoring can help building users to be aware of and promptly fix any deviations in thermal comfort metrics. These measures by themselves will not resolve the issue of potential thermal discomfort, but they certainly raise awareness and are an important first step toward a solution. In addition to having calibrated sensors, the positioning of the sensors plays an important role in the accurate assessment of the thermal environment.

PART 1 MONITOR THERMAL ENVIRONMENT (MAX : 1 PT)

For All Spaces:

1: Thermal comfort monitors

The following requirements are met:

a. The project monitors dry-bulb temperature and relative humidity in occupiable spaces in compliance with the requirements outlined in the Continuous Monitoring Protocols of the Performance Verification Guidebook.

b. The thermal comfort data monitored is made available to occupants through one of the following:

   1. Display screens, with at least one screen located in each 500 m^2 zone of regularly occupied space.
   2. A website or mobile application, with at least one sign located in each 500 m^2 zone of regularly occupied space, indicating where the data may be accessed.

c. The thermal comfort data are updated at least once every 15 minutes.

Note :
Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

REFERENCES

1. Bluyssen PM, Roda C, Mandin C, et al. Self-reported health and comfort in “modern” office buildings: First results from the European OFFICAIR study. Indoor Air. 2016;26(2):298-317. doi:10.1111/ina.12196